Personalized Semantics with User-Context-Aware Calculated Columns

dataMinds Connect 2026 · 60 minutes · Level 300

This session shows how to design user-context-aware calculated columns in Power BI using the Expression Context property and DAX functions such as USERCULTURE(), USERPRINCIPALNAME(), USERNAME(), USEROBJECTID(), and CUSTOMDATA(). We will walk through practical personalization and translation scenarios across supported storage modes, including Direct Lake on OneLake, Import, and DirectQuery. The session also clarifies when calculated columns are evaluated, how materialized versus unmaterialized behaviour affects refresh and query performance.

Flávio Fernandes

Agenda

1. What user-context-aware calculated columns are
2. How they differ from standard calculated columns
3. Demo time
    1. Localization
    2. Report-friendly OLS implementation
    3. Filtered and masked implementation
4. Conclusions and recommendations
5. Q&A

Flávio Fernandes

- CEO of booldata and a Microsoft-certified specialist across the full Fabric and Power BI ecosystem
- More than a decade working at the intersection of data engineering and analytics
- Focused on helping organisations turn ambitious data goals into practical, week-to-week execution
- PhD researcher exploring how companies can responsibly integrate AI into decision-making



Microsoft Certified:
- Fabric Analytics Engineer Associate
- Fabric Data Engineer Associate
- SQL AI Developer Associate
- Power BI Data Analyst Associate

COMPANY

booldata.com

/company/booldata

PERSONAL

flavioapfernandes.com

/in/flavioapfernandes

Thank you, partners















































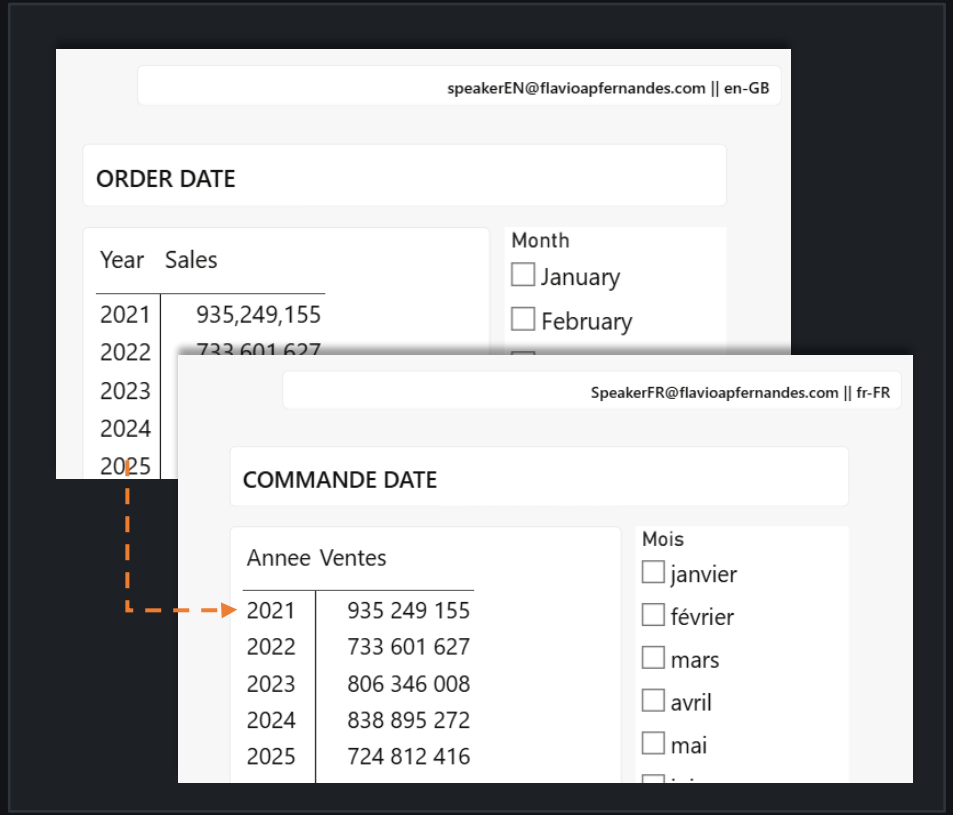


By the End of This Hour

> By the end of this hour, you will know how to build user-context-aware columns in Power BI that dynamically reshape your data based on the active user. You'll walk away understanding how to implement this using both virtual and semi-virtual approaches, without maintaining separate reports or messy row-level security hacks.

A Model That Speaks the User's Language

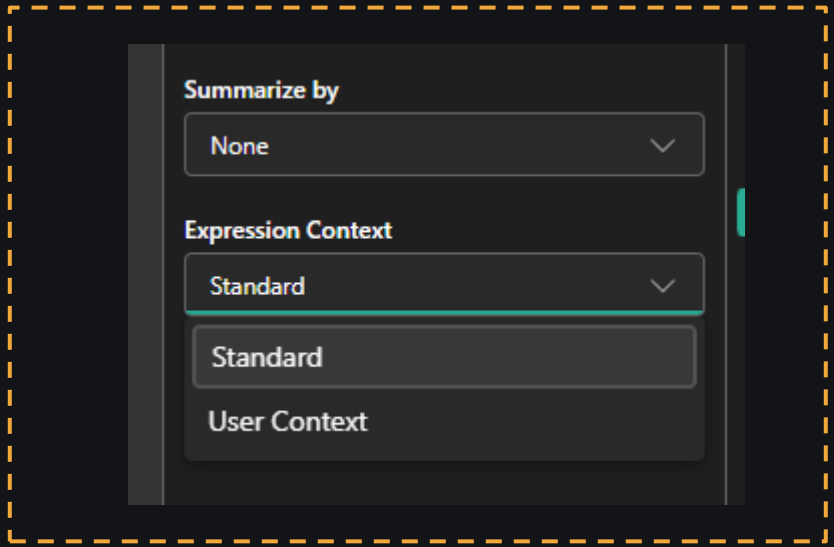
- You already have one semantic model
- Captions may follow the user. The data often does not
- With personalized semantics, it is the same model, but the values belong to the person who opened it



Two Options for Expression Context

Standard
evaluated at refresh

User Context
evaluated at query time
Launched April 2026



What Is a User-Context-Aware Calculated Column?

IS

- Query-time calculated column
- Current user's security perimeter
- Still a column: axis, slicer, filter pane
- Can use:
  - USERCULTURE – Locale
  - USERPRINCIPALNAME, USERNAME, USEROBJECTID - Identity
  - CUSTOMDATA – Connection string

IS NOT

- Metadata translations
- RLS
- OLS
- Field Parameters

Three Differences from a Standard Calculated Column

1. Executed in the current user's security context
2. Result can depend on user identity
3. The column is not materialized*

** Standard calculated columns are materialized only in Import mode*

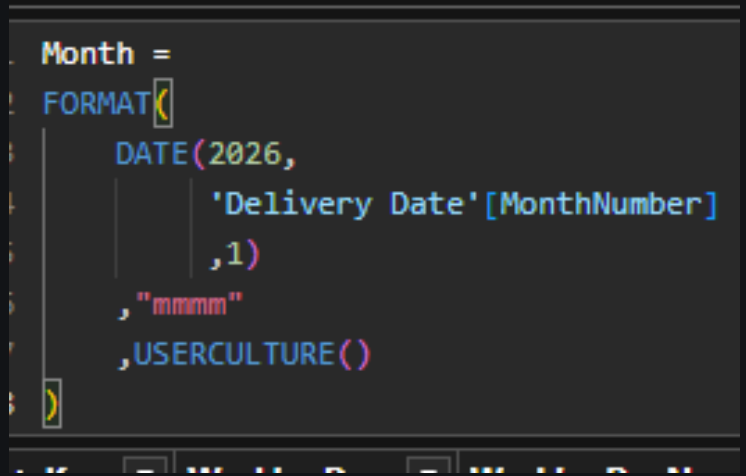
Demo Time

- Import Mode
- Star schema model based on Contoso

Demo 1.1 — Localization - Virtual

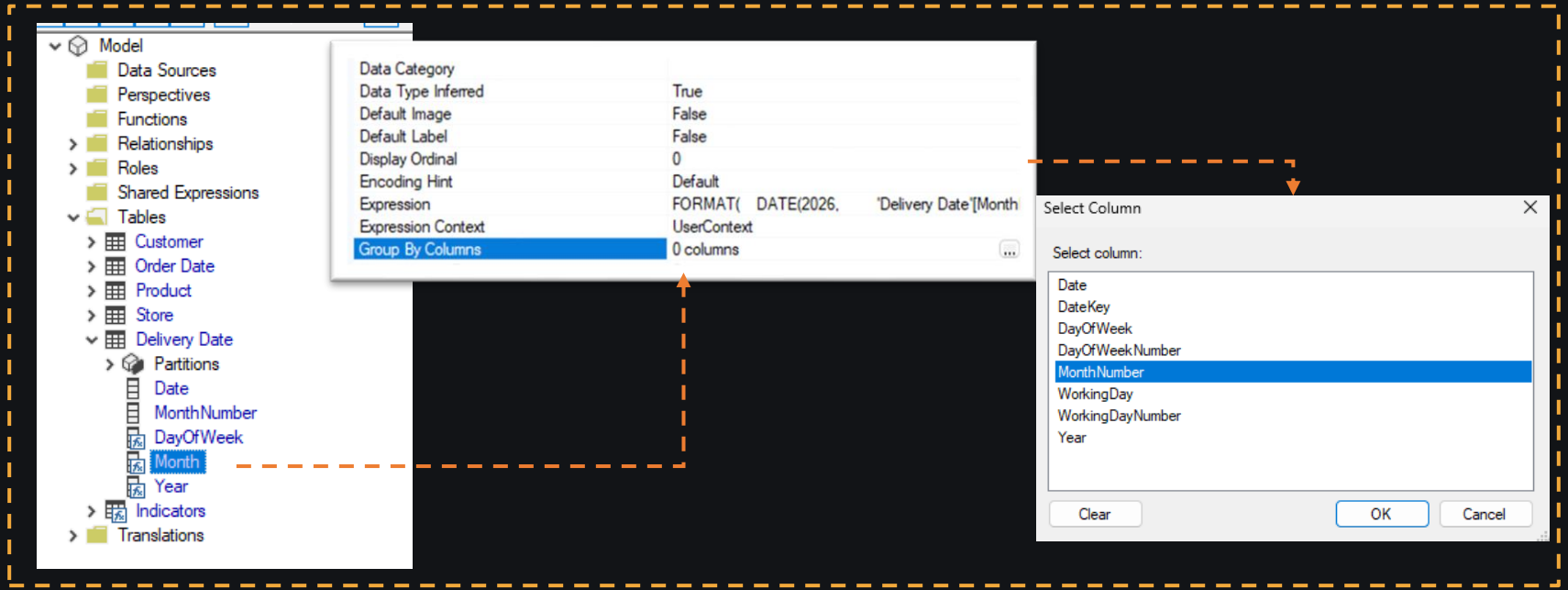
One model. Many languages.

- Standard + USERCULTURE() → error
- Expression Context = User Context
- Depend on Month Number, not Date
- Sort by Column
- Slicer stores the translated string, so bookmarks break
- Group By Columns on Month Number



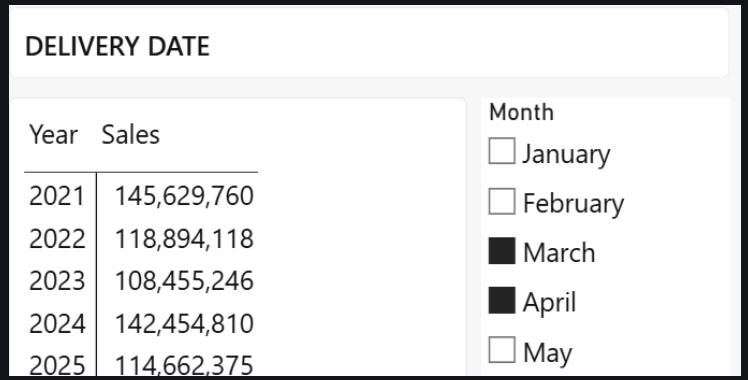
Demo 1.1 — Localization - Virtual

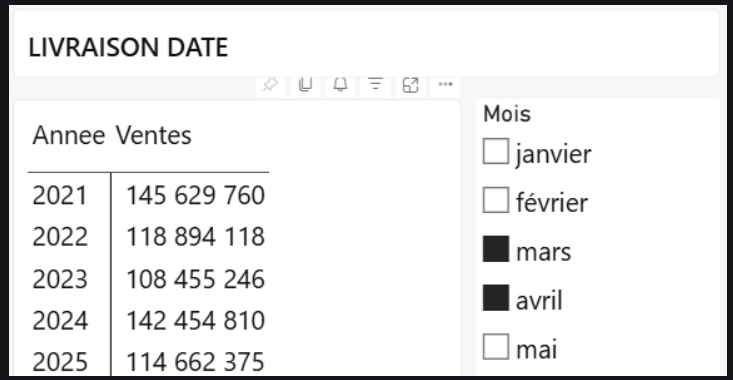
One model. Many languages.



Demo 1.1 — Localization - Virtual

One model. Many languages.





Demo 1.2 — Localization – Semi-Virtual

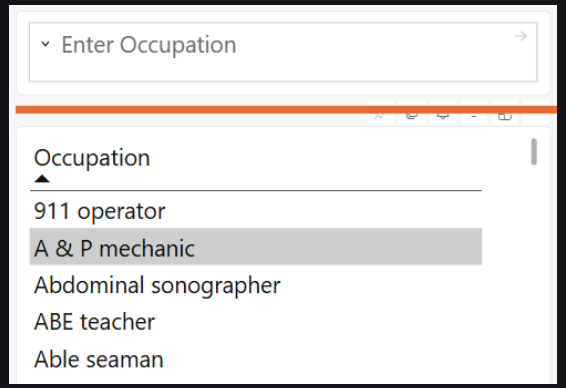
One model. Many languages.

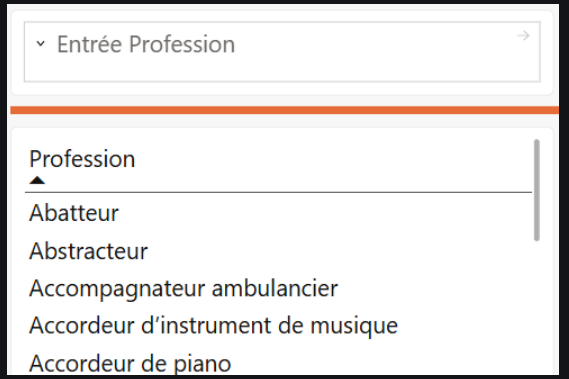
- Standard + USERCULTURE() → error
- Expression Context = User Context
- Different languages are available in the model
- USERCULTURE() defines what the user sees

```
Occupation =
    SWITCH(
        USERCULTURE(),
        "en-US",[Occupation-EN],
        "pt-PT",[Occupation-PT],
        "fr-FR",[Occupation-FR],
        [Occupation-EN]
    )
```

Demo 1.2 — Localization – Semi-Virtual

One model. Many languages.





Demo 2 — Same Report, No Errors

Restricted users see blanks

- OLS hides the column → the visual breaks
- User Context + LOOKUPVALUE → values go blank
- The field still exists in the model
- This is content redaction, not OLS

```
Evaluation =
REPT(
    "★",
    LOOKUPVALUE(
      'Store Sensible Information'[Evaluation]
      ,'Store Sensible Information'[StoreKey]
      ,Store[StoreKey]
    )
)
```

Demo 2 — Same Report, No Errors

Restricted users see blanks

Country	★	★★	★★★	★★★★	★★★★★
Australia	1	2	3		1
Canada	4	1		1	1
France	2	1	2	2	
Germany		1	2	5	2
Italy		1		2	
Netherlands	1	1	1	1	1
Online	1				
United Kingdom	3		3		1
United States	11	2	2	9	3
Total	**23**	**9**	**13**	**20**	**9**

Pais
Total

Demo 3 — Same Report, Masked Data

Users see filtered data + “Others”

- User Context + LOOKUPVALUE → Show data and hide the rest in “Others”
- The field still exists in the model
- This is content redaction, not OLS

```
Category =
VAR _custom = CUSTOMDATA()
VAR _upn = IF ( _custom = BLANK() || _custom = "", USERPRINCIPALNAME(), _custom )
VAR _availableInfo =
    LOOKUPVALUE(
      'Product Sensible Information'[DepartmentName]
      ,'Product Sensible Information'[UserPrincipalName]
      ,_upn
    )
RETURN
    IF(
        ISBLANK(_availableInfo) || PATHCONTAINS(_availableInfo, [DepartmentName]),
        SWITCH(
            USERCULTURE(),
            "en-US",[CategoryName],
            "fr-FR",[CategoryName-FR],
            "pt-PT",[CategoryName-PT],
            [CategoryName]
        ),
        SWITCH(
            USERCULTURE(),
            "en-US","Others",
            "fr-FR","Autres",
            "pt-PT","Outros",
            "Others"
        )
    )
```

Demo 3 — Same Report, Masked Data

Users see filtered data + Others

Department	Sales	Sales % Parent	Sales % Total
Consumer Electronics & Imaging	826,831,435	88%	88%
Home & Living	65,361,706	7%	7%
Entertainment, Culture & Leisure	43,056,013	5%	5%
Total	935,249,155		100%

Departamento	Vendas	Vendas % Superior	Vendas % Total
Casa e Conforto	65.361.706	7%	7%
Outros	869.887.449	93%	93%
Total	935.249.155		100%

Store the ID, Not the Text

- Slicers persist the label unless you stop them
- January ≠ janvier ≠ Janeiro
- Group By Columns → Month Number

FOUR-STEP RECIPE

1. USERCULTURE() in the expression
2. Expression Context = User Context
3. Sort by Column
4. Group By the stable key

Materialization Is Derived

Storage mode	Standard	User Context
Import	Materialized	Unmaterialized
Direct Lake on OneLake	Unmaterialized	Unmaterialized
Direct Lake on SQL	N/A	N/A
DirectQuery	Unmaterialized	Unmaterialized
Dual	Materialized (Import), unmaterialized (DirectQuery)	Unmaterialized
DirectQuery on Power BI semantic models	Unmaterialized	N/A

- Import is the only mode that materializes a Standard calculated column
- Direct Lake on OneLake: always unmaterialized

The Important Performance Warning

GOOD
Simple operators on the same table (storage engine)

BAD
Iterators, complex IF, large table scans, or expensive aggregations

Do not put heavy calculations in a user-context-aware calculated column

Four Things This Cannot Do

1. Relationships
2. Standard calculated columns
3. Calculated tables
4. RLS expressions

Break these rules, and the model will not save.

When to Use Which

User Context

- The value depends on the user
- OR simple same-table arithmetic and you want a virtual column

Standard

- You need a relationship
- The expression is complex
- Query time is more expensive than refresh

You Can Now

- Set Expression Context on purpose
- Ship one multilingual model without breaking the slicer
- Redact a column without breaking the report

Remember: Set User Context and don’t forget Group By

Questions

Personalized Semantics with User-Context-Aware Calculated Columns

Flávio Fernandes

dataMinds Connect 2026

Personalized Semantics with User-Context-Aware calculated columns